Large values of the Gowers-Host-Kra seminorms

Terence Tao · 16 December 2010

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Tanja Eisner and I have just uploaded to the arXiv our paper "Large values of the Gowers-Host-Kra seminorms", submitted to Journal d'Analyse Mathematique. This paper is concerned with the properties of three closely related families of (semi)norms, indexed by a positive integer k:

- The *Gowers uniformity norms* $\|f\|_{U^k(G)}$ of a (bounded, measurable, compactly supported) function $f: G \to \mathbf{C}$ taking values on a locally compact abelian group G, equipped with a Haar measure μ;
- The Gowers uniformity norms $\|f\|_{U^k([N])}$ of a function $f: [N] \to \mathbf{C}$ on a discrete interval $\{1, ..., N\}$; and
- The Gowers-Host-Kra seminorms $\|f\|_{U^k(X)}$ of a function $f \in L^\infty(X)$ on an ergodic measure-preserving system $X = (X, \mathcal{X}, \mu, T)$.

These norms have been discussed in depth in previous blog posts, so I will just quickly review the definition of the first norm here (the other two (semi)norms are defined similarly). The $U^k(G)$ norm is defined recursively by setting

$$\|f\|_{U^1(G)} := \left| \int_G f \, d\mu \right|$$

and

$$\|f\|_{U^k(G)}^{2^k} := \int_G \|\Delta_h f\|_{U^{k-1}(G)}^{2^{k-1}} \, d\mu(h)$$

where $\Delta_h f(x) := f(x+h)\overline{f(x)}$. Equivalently, one has

$$\|f\|_{U^k(G)} := \left(\int_G \dots \int_G \Delta_{h_1} \dots \Delta_{h_k} f(x) \, d\mu(x) d\mu(h_1) \dots d\mu(h_k) \right)^{1/2^k}.$$

Informally, the Gowers uniformity norm $\|f\|_{U^k(G)}$ measures the extent to which (the phase of) f behaves like a polynomial of degree less than k. Indeed, if $\|f\|_{L^\infty(G)} \leq 1$ and G is compact with normalised Haar measure $\mu(G) = 1$, it is not difficult to show that $\|f\|_{U^k(G)}$ is at most 1, with equality if and only if f takes the form $f = e(P) := e^{2\pi i P}$ almost everywhere, where $P: G \to \mathbf{R}/\mathbf{Z}$ is a polynomial of degree less than k (which means that $\partial_{h_1} \dots \partial_{h_k} P(x) = 0$ for all $x, h_1, ..., h_k \in G$).

Our first result is to show that this result is robust, uniformly over all choices of group G:

> **Theorem 1 (L^∞-near extremisers)** Let G be a compact abelian group with normalised Haar measure $\mu(G) = 1$, and let $f \in L^\infty(G)$ be such that $\|f\|_{L^\infty(G)} \leq 1$ and $\|f\|_{U^k(G)} \geq 1 - \epsilon$ for some $\epsilon > 0$ and $k \geq 1$. Then there exists a polynomial $P: G \to \mathbf{R}/\mathbf{Z}$ of degree at most $k-1$ such that $\|f - e(P)\|_{L^1(G)} = o(1)$, where $o(1)$ is bounded by a quantity $c_k(\epsilon)$ that goes to zero as $\epsilon \to 0$ for fixed k.

The quantity $o(1)$ can be described effectively (it is of polynomial size in ϵ), but we did not seek to optimise it here. This result was already known in the case of vector spaces $G = \mathbf{F}_p^n$ over a fixed finite field $\mathbf{F}_p$ (where it is essentially equivalent to the assertion that the property of being a polynomial of degree at most $k-1$ is locally testable); the extension to general groups G turns out to fairly routine.

The basic idea is to use the recursive structure of the Gowers norms, which tells us in particular that if $\|f\|_{U^k(G)}$ is close to one, then $\|\Delta_h f\|_{U^{k-1}(G)}$ is close to one for most h, which by induction implies that $\Delta_h f$ is close to $e(Q_h)$ for some polynomials Q_h of degree at most $k-2$ and for most h. (Actually, it is not difficult to use cocycle equations such as $\Delta_{h+k} f = \Delta_h f \times T^h \Delta_k f$ (when $|f| = 1$) to upgrade "for most h" to "for all h".) To finish the job, one would like to express the Q_h as derivatives $Q_h = \partial_h P$ of a polynomial P of degree at most $k-1$. This turns out to be equivalent to requiring that the Q_h obey the cocycle equation

$$Q_{h+k} = Q_h + T^h Q_k$$

where $T^h F(x) := F(x+h)$ is the translate of F by h. (In the paper, the sign conventions are reversed, so that $T^h F(x) := F(x-h)$, in order to be compatible with ergodic theory notation, but this makes no substantial difference to the arguments or results.) However, one does not quite get this right away; instead, by using some separation properties of polynomials, one can show the weaker statement that

$$Q_{h+k} = Q_h + T^h Q_k + c_{h,k} \tag{1}$$

where the $c_{h,k}$ are small real constants. To eliminate these constants, one exploits the trivial cohomology of the real line. From (1) one soon concludes that the $c_{h,k}$ obey the 2-cocycle equation

$$c_{h,k} + c_{h+k,l} = c_{h,k+l} + c_{k,l}$$

and an averaging argument then shows that $c_{h,k}$ is a 2-coboundary in the sense that

$$c_{h,k} = b_{h+k} - b_h - b_k$$

for some small scalar b_h depending on h. Subtracting b_h from Q_h then gives the claim.

Similar results and arguments also hold for the $U^k([N])$ and $U^k(X)$ norms, which we will not detail here.

Dimensional analysis reveals that the L^∞ norm is not actually the most natural norm with which to compare the U^k norms against. An application of Young's convolution inequality in fact reveals that one has the inequality

$$\|f\|_{U^k(G)} \leq \|f\|_{L^{p_k}(G)} \tag{2}$$

where p_k is the critical exponent $p_k := 2^k/(k+1)$, without any compactness or normalisation hypothesis on the group G and the Haar measure μ. This allows us to extend the $U^k(G)$ norm to all of $L^{p_k}(G)$. There is then a stronger inverse theorem available:

> **Theorem 2 (L^{p_k}-near extremisers)** Let G be a locally compact abelian group, and let $f \in L^{p_k}(G)$ be such that $\|f\|_{L^{p_k}(G)} \leq 1$ and $\|f\|_{U^k(G)} \geq 1-\epsilon$ for some $\epsilon > 0$ and $k \geq 1$. Then there exists a coset H of a compact open subgroup H of G, and a polynomial $P : Hto\mathbf{R}/\mathbf{Z}$ of degree at most $k-1$ such that $\|f - e(P)1_H\|_{L^{p_k}(G)} = o(1)$.

Conversely, it is not difficult to show that equality in (2) is attained when f takes the form $e(P)1_H$ as above. The main idea of proof is to use an inverse theorem for Young's inequality due to Fournier to reduce matters to the L^∞ case that was already established. An analogous result is also obtained for the $U^k(X)$ norm on an ergodic system; but for technical reasons, the methods do not seem to apply easily to the $U^k([N])$ norm. (This norm is essentially equivalent to the $U^k(\mathbf{Z}/\tilde{N}\mathbf{Z})$ norm up to constants, with $\tilde{N}$ comparable to N, but when working with near-extremisers, norms that are only equivalent up to constants can have quite different near-extremal behaviour.)

In the case when G is a Euclidean group $\mathbf{R}^d$, it is possible to use the sharp Young inequality of Beckner and of Brascamp-Lieb to improve (2) somewhat. For instance, when $k=3$, one has

$$\|f\|_{U^3(\mathbf{R}^d)} \leq 2^{-d/8} \|f\|_{L^2(\mathbf{R}^d)}$$

with equality attained if and only if f is a gaussian modulated by a quadratic polynomial phase. This additional gain of $2^{-d/8}$ allows one to pinpoint the threshold $1-\epsilon$ for the previous near-extremiser results in the case of U^3 norms. For instance, by using the Host-Kra machinery of characteristic factors for the $U^3(X)$ norm, combined with an explicit and concrete analysis of the 2-step nilsystems generated by that machinery, we can show that

$$\|f\|_{U^3(X)} \leq 2^{-1/8}\|f\|_{L^2(X)}$$

whenever X is a *totally* ergodic system and f is orthogonal to all linear and quadratic eigenfunctions (which would otherwise form immediate counterexamples to the above inequality), with the factor $2^{-1/8}$ being best possible. We can also establish analogous results for the $U^3([N])$ and $U^3(\mathbf{Z}/N\mathbf{Z})$ norms (using the inverse U^3 theorem of Ben Green and myself, in place of the Host-Kra machinery), although it is not clear to us whether the $2^{-1/8}$ threshold remains best possible in this case.

Comments (3)



Anon · 5 October 2018

Dear Terence Tao, could you please indicate some more detail on how one deduces from (1) the 2-cocyle equation c_{h,k} + c_{h+k,l} = c_{h,k+l} + c_{k,l} ? In your paper with Eisner I cannot see any details either for how to do this deduction.



Terence Tao · 5 October 2018

From (1) we have $Q_{h+k+l} = Q_h + T^h Q_{k+l} + c_{h,k+l}$, $Q_{k+l} = Q_k + T^k Q_l + c_{k,l}$, $Q_{h+k+l} = Q_{h+k} + T^{h+k} Q_l + c_{h+k,l}$, and $Q_{h+k} = Q_h + T^h Q_k + c_{h,k}$. Use these identities to expand out Q_{h+k+l} in two different ways and collect terms.



Anonymous · 6 October 2018

Great, thank you!